

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**UNIT TEST****14-06-2020****JEE(Main+Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II & III]****PAPER-1****Time : 3 Hours****Maximum Marks : 180****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

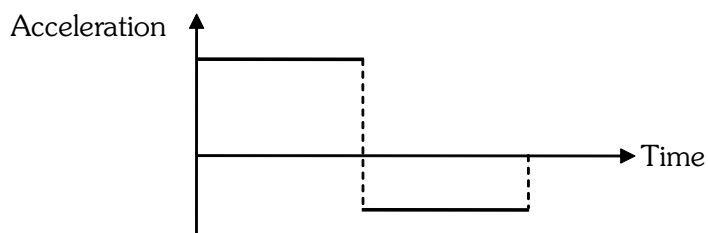
SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If the speed v of a particle of mass m as function of time t is given by $v = \omega A \sin \left[\left(\sqrt{\frac{k}{m}} \right) t \right]$, where A has dimension of length. Which of the following statement(s) is/are **INCORRECT** ?
- (A) the argument of trigonometric function must be a dimensionless quantity
- (B) Dimensional formula of ω is LT^{-1}
- (C) Dimensional formula of k is MLT^{-2}
- (D) Dimensional formula of $\sqrt{\frac{k}{m}}$ is T

Space for Rough Work

2. In the given figure is shown, how acceleration of particle moving along the x-axis varies in the first and the second time intervals of equal durations. If the particle starts from rest, which of the following conclusions can you make?



- (A) In the first half time, it speeds up and in the second half time, it slows down.
 (B) In the first half time, it covers more distance than it covers in the second half time.
 (C) In the first half time, it covers lesser distance than it covers in the second half time.
 (D) During the first half time, it moves in the positive x-direction and in the second half time, it moves in the negative x-direction.
3. A particle is moving in a straight line for a finite non-zero time interval.
- (A) Average speed is greater than or equal to magnitude of average velocity.
 (B) Average speed is less than or equal to magnitude of average velocity.
 (C) Average speed may be zero.
 (D) Magnitude of average velocity may be zero.

Space for Rough Work

4. A particle moving along a straight line with uniform acceleration has velocities 7 m/s at P and 17 m/s at Q. R is the mid point of PQ. Then-
- (A) the average velocity between R and Q is 15 m/s
 - (B) the ratio of time to go from P to R and that from R to Q is 3 : 2
 - (C) the velocity at R is 10 m/s
 - (D) the average velocity between P and R is 10 m/s
5. A freely falling body travelled 35 m in last second of its journey :-
- (A) The height of the point of release from ground is 80 m
 - (B) The distance travelled in the first second is 5 m
 - (C) The maximum speed attained by the body is 40 m/s
 - (D) The maximum speed attained by the body is 20 m/s
6. A particle is moving in x-y plane such that its position at any time is given by $x = (2t + 1)$ m $y = (t^3 - t^2 + 1)$ m. Then select the correct option :-
- (A) Initial speed of particle is 2m/s.
 - (B) Acceleration of particle at 3 sec is 16 m/s²
 - (C) Motion of particle is uniformly accelerated.
 - (D) Direction of velocity of particle at 1 sec is making angle $\tan^{-1}\left(\frac{1}{2}\right)$ from x-axis.

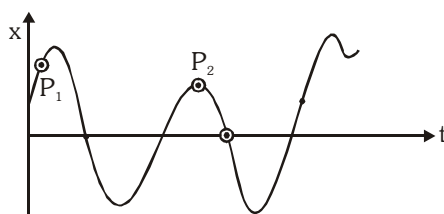
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

A particle is moving along x-axis under a force such that its position-time graph is as shown in figure.



7. As the particle passes position P_1
 - (A) it is moving along negative x-direction with increasing speed
 - (B) it is moving along positive x-direction with decreasing speed
 - (C) it is moving along negative x-direction with decreasing speed
 - (D) it is moving along negative x-direction with constant speed
8. As the particle passes position P_2
 - (A) it is moving along negative x-direction with maximum speed
 - (B) it is moving along positive x-direction with minimum speed
 - (C) it is instantaneously at rest and will now move along positive x-direction
 - (D) it is instantaneously at rest and will now move along negative x-direction

Space for Rough Work

Paragraph for Questions 9 and 10

If $x = 3t^2 - 2t$ is position of a particle moving along x-axis.

9. Velocity of particle is given by :-

(A) $6t + 2$

(B) $3t - 2$

(C) $2(3t - 1)$

(D) $2t - 3t^2$

10. Minimum speed of particle is :-

(A) 6

(B) 2

(C) 4

(D) 0

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

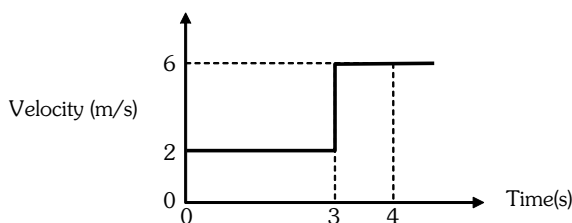
- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. In the following graph variation with time (t), in velocity (v) of a particle moving rectilinearly is shown. What is average velocity in m/s of the particle in time interval from 0 s to 4 s?



2. A particle starts from rest and its acceleration at any time t is given by $a = (3 - 0.75 t^2) \text{ m/s}^2$. Find its maximum velocity (in m/s).

Space for Rough Work

3. A car starting from rest is accelerated at constant rate until it attains a speed of 12 m/s. It moves with the constant speed 12 m/s for some time and then it is retarded at a constant rate until it comes to rest. Considering that the car moves with constant speed for half of the time of total journey, what will be the average speed of the car for the whole journey (in m/s) ?
4. A particle is thrown upward with a velocity of 50 m/s. It passes the point A at 3 sec and 7 sec after the projection. If height of point A (in m) from the point of projection is h. Find $\frac{h}{100}$.

Space for Rough Work

5. A particle moves along a straight line. The acceleration of the particle as function of time is given by $a = (6t - 12) \text{ m/s}^2$. Initial velocity of particle is 9 m/s . If distance traveled by particle in 5 sec is s (in m). Find $\frac{s}{5}$.
6. One calorie is a unit of heat energy and its value is 4.18 J where $1 \text{ J} = 1 \text{ kg-m}^2/\text{sec}^2$. Suppose we use a new system of units in which the unit of mass equals 2 kg , the unit of length equals 5 m and the unit of the time is 10 sec . Then find the value of one calorie in the new system of units.

Space for Rough Work

7. If velocity (V), force (F) and time (T) are taken to be fundamental quantities. If the dimensional formula of energy (Q) is represented by the $Q = kV^x F^y T^z$. Then fill the $\frac{(x+y+z)}{2}$ in OMR sheet.
8. The maximum and minimum value of resultant of two forces acting on an object are 15 and 3N respectively. If each force is increased by 1N and angle between the forces is made 90° then resultant becomes $\sqrt{x}N$. Find $\frac{x}{100}$.

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Which of the following order is/are **CORRECT** for indicated property?
(A) $O > N > C$ (Z_{eff} value)
(B) $P < S < Cl$ (number of protons)
(C) s-subshell < p-subshell < d-subshell (value of azimuthal quantum number)
(D) 1st period < 2nd period < 3rd period (number of s-block elements)
 2. Which of the following pair(s) of species having same 'spin only' magnetic moment in ground state electronic configuration?
(A) Cr, Cu (B) Sc, Cu (C) Na, Rb (D) Zn, Cd
 3. Which of the following statement is/are **CORRECT** about lother meyer curve?
(A) Representative elements are present at bottom of curve
(B) The elements present at the peak (maxima) are chemically very reactive
(C) It is a curve between atomic volume and atomic density
(D) Alkaline earth metal present at descending position of curve
-

Space for Rough Work

4. Which of the following option/s **CORRECTLY** match group number and period number for given atomic number (Z)?
- (A) $Z = 56 \rightarrow$ group number = 2, period number = 6
(B) $Z = 78 \rightarrow$ group number = 10, period number = 6
(C) $Z = 53 \rightarrow$ group number = 7, period number = 6
(D) $Z = 40 \rightarrow$ group number = 4, period number = 5
5. Which of the following is/are **CORRECT** order for property mentioned in bracket?
- (A) $\text{Si} < \text{P} < \text{Cl}$ (second ionisation energy)
(B) $\text{Cl} > \text{F} > \text{Br}$ (electron affinity)
(C) $\text{Cl} < \text{Mg} < \text{Ca}$ (atomic size)
(D) $\text{Zn} > \text{Cd} > \text{Hg}$ (first ionisation energy)
6. Select the species in which coordinate bond is present
- (A) O_3 (B) CCl_4 (C) BF_4^- (D) NH_4^+
-

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Amount of energy required to remove the most loosely bonded electron from an isolated gaseous atom from its ground state electronic configuration is called ionisation energy. The value of ionisation energy mainly depends on (i) Z_{eff} (ii) Atomic radius (iii) Screening effect of inner electron. Successive ionisation energy of element always follow increasing order.

7. Ionisation energy of 'K' would be same as :
(A) electron affinity of K^+ (B) electronegativity of K^+
(C) electron affinity of K (D) Ionisation energy of Ca^+
8. **INCORRECT** order of ionisation energy is :
(A) $Pb > Sn$ (B) $Na^+ > Mg^+$ (C) $Li^+ < O^+$ (D) $Be^+ < C^+$

Paragraph for Questions 9 and 10

In long form of periodic table elements are classified into four blocks s, p, d and f based on their electronic configuration. Periodic properties and other physical properties usually show a regular pattern of change within a group or along period with some irregularities.

9. On moving from Li to F in the second period, there would be a regular decreases in :
(A) Non metallic nature (B) Atomic radius
(C) Ionisation energy (D) Electronegativity
10. In which of the following set of elements first element having higher electronegativity than second element on Pauling scale?
(A) Ga, Al (B) S, O (C) P, N (D) B, C

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- According to Slater's rule, calculate effective nuclear charge (Z_{eff}) on last electron of F^- .
- Find out total number of combination of atomic orbitals, which form π -bond along y axis.

(i) $p_y + p_y$	(ii) $d_{xy} + d_{xy}$	(iii) $p_x + p_x$	(iv) $d_{xy} + p_x$
(v) $d_{xy} + d_{x^2-y^2}$	(vi) $d_{xz} + d_{xy}$	(vii) $p_z + d_{xz}$	

Space for Rough Work

3. Calculate $\frac{(x+y)}{2}$ for given species BF_3 , BeF_2 , AlCl_3 , NH_3 , PCl_5 , SF_6 , CCl_4 , IF_7 , CO_2 and XeF_4 where
x = Number of hypovalent species
y = Number of hypervalent species
4. Calculate $\frac{(x+y)}{2}$:
where
x = Maximum number of degenerate orbitals in 2nd excited state of Li^{2+}
y = Electronegativity value of most electronegative element on Pauling scale
5. Find maximum number of unpaired electrons are possible in any of the 4th period element in its ground state electronic configuration.

Space for Rough Work

6. Find total number of third period element having less ionisation energy than the element of next atomic number.
7. If electronegativity of an element is 3.5 on Pauling scale and ionisation energy is 15 eV/atom then find electron affinity of that element in eV/atom.
8. In how many of following pairs, second species has higher ionisation energy than first species :
- | | | | |
|--------------------------------------|----------------------|--------------|---------------------------|
| (i) Na^+ , Mg^{2+} | (ii) S, Cl | (iii) Cu, Zn | (iv) Xe, Kr |
| (v) B, Be | (vi) O, O^+ | (vii) Al, Si | (viii) Cl^- , Cl |

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. If the range of the expression $\frac{x^2+2x-11}{x-3}$ for real values of x is $(-\infty, a] \cup [b, \infty)$ then which of the following is/are true -
- (A) $\sqrt{a^2+b^2}=13$ (B) $\sqrt{a+b}=a$
- (C) a and b are coprime numbers (D) $\log_{b-a}(b+a) = \frac{a^2}{b}$
2. $(a-2)x^2 + 2\sqrt{3}x + a > 0$ for all real value of x , then -
- (A) $a < -1$ (B) $a > 3$ (C) $a > 4$ (D) $3 < a < 8$
-

Space for Rough Work

3. The solution(s) of the equation, $\log_{10}^2 x + \log_{10} x^2 = \log_{10}^2(2) - 1$ is/are -
 (A) $1/10$ (B) $1/20$ (C) $1/2$ (D) $1/5$
4. Which of the following option(s) is(are) true
 (A) sum of all roots of the equation $|9 - x| = 5$ is 4
 (B) sum of all roots of the equation $|9 - x| = 5$ is 18
 (C) sum of all roots of the equation $|x - 9| = 5$ is 18
 (D) sum of all roots of the equation $|x - 9| = 5$ is 14
5. If x and y are integers such that $y + 2x + 3 - xy = 0$, then which of the following option(s) is(are) true
 (A) maximum value of $x + y$ is 6 (B) minimum value of $x + y$ is -6
 (C) maximum value of $x + y$ is 9 (D) minimum value of $x + y$ is -3
6. If $f(x)$ is monic polynomial of degree 3 such that $f(0) = 0$, $f(1) = 1$, $f(2) = 2$ then which of the following option(s) is/are true ?
 (A) $f(4) = 4$ (B) $|f(-1)| = 1$ (C) $f(-1) = -7$ (D) $f(-2) = -26$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Let $f(x) = 4x^2 - 4ax + a^2 - 2a + 2$ be a quadratic polynomial in x and 'a' be a real number.

On the basis of above information, answer the following :

7. If x-coordinate of the vertex of the graph of $y = f(x)$ is less than 0 and for $x \in [0, 2]$, the minimum value of $f(x)$ is 10, then the set of all values of 'a' is -
 (A) $\{-2, 4\}$ (B) $\{4\}$ (C) $\{-2\}$ (D) $\{1\}$
8. If for $x \in [0, 2]$ minimum value of $f(x)$ is -7 and x-coordinate of the vertex of the graph of $y = f(x)$ is greater than 2, then value of 'a' is -
 (A) 2 (B) 3 (C) 4 (D) 5

Paragraph for Questions 9 and 10

If $f(x) = |x - 3| + 2|x - 1|$ and $g(x) = -2x^2 + 3x - 1$.

On the basis of above information, answer the following questions :

9. Minimum value of $f(x)$ is -
 (A) 0 (B) 1 (C) 2 (D) 5
10. Number of solution(s) of the equation $f(x) = g(x)$ is -
 (A) 0 (B) 1 (C) 2 (D) 3

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		-			
●	●	0	0	●	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the equation $x^4 + (k + 1)x^3 + (k + 2)x^2 + (k + 1)x + 1 = 0$ has no real roots, then number of integral values of k is
2. For $8|x^2 + k| = 3|x|$ to have real solution, the maximum value of k is $\frac{\lambda_1}{\lambda_2}$, where λ_1 & λ_2 are relatively prime, then $\sqrt{\lambda_2} - \lambda_1$ is equal to
3. Let $a_1, a_2, a_3, \dots$ are in A.P. If $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2 + p}{q^2 + q}$, $p \neq q$, then the value of $\frac{a_5}{a_{15}}$ is
4. If $xy + yz = 18$, $yz + zx = 20$ and $zx + xy = 8$, then maximum value of $|x| + |y| + |z|$ is

Space for Rough Work

5. If $x = \frac{\sqrt{2}+1}{\sqrt{2}-1}$, $y = \frac{\sqrt{2}-1}{\sqrt{2}+1}$ then the value of $\frac{x^3+y^3}{10}$ is
6. $x^2 + bx + c = 0$, where b and c are rational numbers. If $3-\sqrt{3}$ is one root of the equation, then the value of $\left|\frac{b}{2c}\right|$ is
7. If α is root of $\sqrt{x+3} = -2x$ then the value of $|\alpha|$ is
8. If $(x^2 - 1)^3 - (2x - 1)^3 = x^3(x - 2)^3$, then sum of all roots of the equation is
-

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator